VNR Vignana Jyothi Institute of Engineering

Computer Science and Engineering | Course Code: CS303

Computer Networks & Internet Protocols - Comprehensive Syllabus Lecture

Syllabus Focus: CS303 Full Syllabus Comprehensive Lecture Notes Units 1-5

Summary: Complete syllabus lecture notes covering Computer Networks & Internet Protocols. Inclu

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Physical & Data Link Protocol Foundations

" Syllabus Scope: Transmission media, framing, error detection CRC-32, Hamming code, sliding window flow

Unit 2: Medium Access & Local Area Networks

" Syllabus Scope: ALOHA, CSMA/CD Ethernet, CSMA/CA WiFi 802.11, VLANs, STP Spanning Tree Protocol, ARP pr

CS303: Computer Networks & Internet Protocols - Advanced Topics & Assessment

Academic Track: Computer Science and Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Network Layer & Routing Protocols

" IPv4 and IPv6 addressing, CIDR subnetting, Distance Vector Bellman-Ford, Link State Dijkstra/OSPF, Border Ga

Unit 4: Transport Layer & Congestion Control

" UDP datagrams, TCP 3-way handshake, connection teardown, TCP flow control, AIMD congestion control Slow Star

Unit 5: Application Layer Protocols & Network Security

" DNS resolution hierarchy, HTTP/1.1 vs HTTP/2 vs HTTP/3 QUIC, SMTP, TLS/SSL handshake, firewalls and NAT

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

1. Subnet an IP block 192.168.10.0/24 into 4 subnets with varying host allocations, stating network and b
2. Explain TCP Tahoe versus TCP Reno congestion avoidance, drawing the congestion window size over round-
3. Calculate the CRC checksum for data frame 1101011011 using generator polynomial Gx = x^4 + x + 1.

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* James F. Kurose, Keith W. Ross - Computer Networking: A Top-Down Approach Pearson 8th Ed
* Andrew S. Tanenbaum, David J. Wetherall - Computer Networks Pearson 5th Ed

Authorized NoteVault Educational Document " Verified for Registered University Students